

Invariant Structural Value: Fundamental Constants and Formulas as Invariant Relations

Rowan Brad Quni-Gudzinis

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Abstract

We develop the thesis that the fundamental constants and formulas of physics encode invariant relations rather than unit-dependent magnitudes. Measurable physics is characterized as the invariant quotient of a larger non-measurable mathematical structure modulo redundancy groups: units, coordinates, gauge, basis, and scale. The exponential constant e and the circle constant π are shown to be fixed points of two canonical self-referential operations on a primitive mark-and-distinction: e of self-application and π of self-closure. The Euler identity is the joint fixed point of these operations. Every claim carries an explicit disconfirmation condition.

Introduction

A measured physical quantity is a number that survives all arbitrary choices. The unit system, the coordinate system, the gauge choice, the Hilbert-space basis, and the energy scale are all choices; what is invariant under them is what is physical. This essay develops the structuralist reading that follows: fundamental constants and formulas encode **invariant relations** — the logical architecture of physical law — rather than magnitudes in human units.

Three claims are developed. First, the invariant content of a constant or formula is a dimensionless ratio, a symmetry datum, a topological index, or a fixed-point value. Second, quantum mechanics is the canonical example: measurable content is the projective ray, the spectrum, the transition amplitude, and the quantized invariant, extracted from a non-measurable mathematical total space by quotients under redundancy groups. Third, the constants e and π arise as the fixed points of self-reference itself: e of self-application, π of self-closure, with the Euler identity $e^{i\pi} + 1 = 0$ as their joint fixed point.

Measured Quantities as Invariants

Dimensionless ratios and unit bridges

Dimensionful constants are best understood as **bridges between categories of quantity**. The speed of light c identifies space with time; its invariant content is the null cone and the Minkowski metric signature, not the number $3 \times 10^8 \text{ m s}^{-1}$ in human units. Planck's constant $\hbar$ identifies energy with frequency and action with phase; its invariant content is unitarity and superposition. Newton's constant G identifies mass-energy with spacetime curvature; its invariant content is that gravity is geometry.

Setting $c = \hbar = G = k_B = 1$ merely chooses units. What remains after the bridges are normalized is the network of dimensionless relations: mass ratios m_p/m_e , coupling strengths such as the fine-structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c)$, mixing angles and phases, and topological indices. These are the only numbers that can be compared across unit systems, coordinate systems, and theories.

Structural markers, not magnitudes

The invariant structural value of a fundamental number is its **place in the network of lawful relations**. A coupling strength is not a decimal but the weight of a vertex in a quantum field theory. A mass ratio is not a kilogram value but a hierarchy between scales. A mixing angle is not an arcminute but a rotation between flavor and mass bases. A Chern number is not a conductance reading but a quantized invariant of a bundle.

This reframing makes precise what the decimal values obscure: the physical content is the relation, and the magnitude in any unit system is a projection of that relation onto a coordinate system.

Quantum Mechanics as Invariant Structure

Projective rays and spectra

In quantum mechanics, the measurable content is not the wavefunction but its **ray** — the equivalence class under global phase. Born probabilities

$$P(i) = |\langle i|\psi\rangle|^2$$

are invariant under phase changes $|\psi\rangle \rightarrow e^{i\theta}|\psi\rangle$ and under unitary basis changes. Observables are spectra of self-adjoint operators; the spectrum is invariant under $O \rightarrow UOU^\dagger$. The physical state is therefore $[\psi] = \{e^{i\theta}|\psi\rangle\}$ in projective Hilbert space, and a measured energy is a spectral invariant.

S-matrix and topological invariants

Scattering content is the S-matrix, invariant under field redefinitions, gauge choices, and renormalization scheme. Many measurable quantum numbers are topological: the Aharonov–Bohm phase is a holonomy, the quantum Hall conductance is a first Chern number, the Berry phase is an integral of curvature over a parameter cycle. These are quantized invariants of a mathematical structure, not continuously tunable decimals. [TERRITORY — the claim that measurable content is invariants of a mathematical structure is falsifiable: a gauge-dependent observable that is nonetheless measured would disconfirm it.]

The Non-Measurable Mathematical Scaffolding

The invariant content is extracted from a larger mathematical structure containing elements that are **not directly measurable**: the complex phase and wavefunction, gauge potentials and fiber-bundle connections, path-integral histories, ghost fields and BRST cohomology, bare parameters and infinite renormalization constants, and complexified kinematic spaces.

These are not decorations; they are the total space whose invariants are the measured world. In gauge theory, unphysical degrees of freedom are included and the physical Hilbert space is the BRST cohomology — the subspace annihilated by the BRST charge modulo exact states. In renormalization, bare parameters are scheme-dependent and unphysical while observables are finite and invariant. The relationship is:

Physical observables = invariants of a larger mathematical structure / redundancy groups.

Non-measurable object	Measurable invariant
wavefunction phase $e^{i\theta}$	relative phase, interference
gauge potential A_μ	Wilson loop, field strength $F_{\mu\nu}$
path-integral histories	S-matrix elements
ghost fields	BRST cohomology classes
bare parameters	renormalized couplings
complex momentum plane	poles and residues

The imaginary, non-measurable mathematics is the coordinate system in which physical invariants become computable; measurement is the extraction of the invariant after all arbitrary choices are removed.

Self-Adjointness and Self-Reference: e and pi from Mark and Distinction

The primitive distinction

A mark is a boundary with an inside and an outside; drawing a distinction creates the first pair **0** and **1**. Self-reference occurs when the mark re-enters its own field, or when a form is applied to itself. Self-adjointness is the mirror form of self-reference: an operator equal to its own adjoint,

$$A = A^\dagger,$$

is the fixed point of the adjoint involution, and its spectrum is real. The two canonical ways a distinction can become self-consistent are self-application and self-closure.

e as the invariant of self-application

The exponential constant **e** is the fixed point of the differential equation

$$\frac{d}{dx} f(x) = f(x),$$

the form whose rate of change equals itself; the normalized solution is $f(x) = e^x$. Equivalently, repeated self-application of a small distinction yields

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

This is a feedback loop: take the current state, add a fraction of itself, repeat. The fixed point of that loop is **e**. [TERRITORY — disconfirmation: if **e** can be exhibited as requiring an additional primitive beyond self-application of a mark-and-distinction (a resource not derivable from the calculus of indications), the claim is false.]

pi as the invariant of self-closure

The circle is the simplest distinction that closes on itself. Its circumference to diameter ratio is scale-invariant: this is the period of self-enclosure. In spectral terms, the self-adjoint momentum operator on a circle,

$$\hat{p} = -i \frac{d}{d\theta},$$

with periodic boundary conditions has eigenfunctions $e^{in\theta}$; the consistency condition $e^{in(\theta+2\pi)} = e^{in\theta}$ forces the period 2π . Thus π is the invariant that makes self-adjoint differentiation on a compact domain consistent. [TERRITORY — disconfirmation: if a self-referential equation is exhibited whose fixed point is a constant other than **e** or π with no structural characterization, the fixed-point reading is incomplete.]

Euler identity as joint fixed point

The identity

$$e^{i\pi} + 1 = 0$$

is the structural relation between self-application and self-closure. The constant e is the fixed point of growth; π is the fixed point of closure; i is the fixed point of self-negation, since $i^2 = -1$; -1 is the distinction itself. Exponentiating imaginary growth by the half-period yields negation. Growth, rotation, and self-reference compose into the elementary logical operation of distinction.

Compact closed structure

Certain physical theories are compact closed: every object has a dual, and every process can be bent back into itself, forming traces. For the compact group $U(1)$, the exponential map is $\theta \mapsto e^{i\theta}$ with kernel $2\pi\mathbb{Z}$; hence $e^{i\pi} = -1$. Here e is the base of the exponential map and π is the half-period of the compact group. Compact closure is what allows the feedback loop to exist, and e and π are the invariants of that loop.

Formal derivation (constructive, pre-registered)

C3 is claimed constructively, not by pattern-matching. Each constant is exhibited as the unique fixed point of a specified self-referential equation over the calculus of indications.

e as the fixed point of self-application. Let T be the operator that maps a function to its own rate of change: $T[f] = f'$. The fixed-point equation

$$T[f] = f, \quad f(0) = 1$$

has the unique analytic solution $f(x) = e^x$, hence $e = f(1)$. The construction is explicit: the Picard iteration

$$f_0 = 1, \quad f_{n+1}(x) = 1 + \int_0^x f_n(t) dt$$

converges uniformly on compacta to f , yielding the series $e = \sum_{n=0}^{\infty} 1/n!$. Equivalently, the iteration of a small distinction onto itself, $a_n = (1 + 1/n)^n$, converges monotonically to e . The primitive is the distinction between a form and its own increment: self-application is the operation that feeds a form back into itself.

π as the fixed point of self-closure. Closure is the requirement that a form return to itself. The self-adjoint momentum operator on a circle, $\hat{p} = -i d/d\theta$, is self-adjoint only for periodic boundary conditions $f(\theta + 2\pi) = f(\theta)$; its eigenfunctions are $e^{in\theta}$, and the periodicity condition forces the fundamental period to be 2π . Equivalently, the exponential map $\exp : \mathbb{R} \rightarrow U(1)$, $x \mapsto e^{ix}$, has kernel $2\pi\mathbb{Z}$: self-closure of the exponential map fixes the period, and π is its half-period, the least positive x with $e^{ix} = -1$. The primitive is the distinction between a line and its return: self-closure is the operation that identifies a path with its endpoint.

Euler identity as the joint fixed point. The two fixed points compose: $e^{i\pi} = -1$ states that self-application iterated through the imaginary half-period of self-closure yields the elementary distinction -1 . The identity $e^{i\pi} + 1 = 0$ re-states the original mark pair $0, 1$.

Constructive verification. Each step above is an exhibited equation with a unique solution; no numerical value is assumed. Independent recomputation of the fixed-point values e , π , and the identity $e^{i\pi} + 1 = 0$ is recorded in the fit-verify artifact. [TERRITORY — the derivation is disconfirmed if either constant can be shown to require an additional primitive not derivable from distinction and self-reference, or if a self-referential equation is exhibited whose fixed point is a constant other than e or π with no structural characterization.]

Falsifiability and Symmetric Audit

Disconfirmation conditions

C1. Disconfirmed if a dimensionful constant is shown to carry invariant physical content beyond its role as a unit bridge, or if a claimed invariant is demonstrated to be scale-dependent where asserted scale-invariant.

C2. Disconfirmed if a measurable quantity is exhibited that is not expressible as an invariant under the enumerated redundancy groups — for example, a gauge-dependent observable that is nonetheless measured.

C3. Disconfirmed if e or π can be shown to require input beyond mark-and-distinction plus self-reference, or if a self-referential equation is exhibited whose fixed point is a constant other than e or π with no structural characterization.

Symmetric audit

The incumbent accounts — structural realism, duality-based accounts of theoretical equivalence, relational quantum mechanics, and the gauge-theoretic treatment of redundancy — are graded with the same kill-criteria as the present framework. The Standard Model's measured parameters and the underdetermination arguments against structural realism are audited with identical rigor, not assumed superior.

Conclusion

Fundamental constants and formulas encode invariant relations: dimensionless ratios, symmetry data, topological indices, and fixed-point values. Measurable physics is the invariant quotient of a larger non-measurable mathematical structure under redundancy groups. The constants e and π are the fixed points of self-reference itself — self-application and self-closure on a primitive distinction — and the Euler identity is their joint fixed point. The decimal magnitudes in human units are shadows of these invariants; the invariant structural value is the place in the network of lawful relations.

Declarations

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